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Behavior of two-dimensional C/SiC composites subjected to thermal cycling in controlled environments

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Abstract

Thermal fatigue behavior of two-dimensional carbon fiber reinforced SiC matrix composites fabricated by chemical vapor infiltration technique was investigated using an on-line quench method in controlled environments which simulated an aero-engine gas. A system of damage information acquisition (SDIA) was developed to study changes in electrical resistance of the C/SiC composites during their damage in dynamic testing. Damage to composites was assessed by the ultimate tensile strength (UTS) and SEM characterization. The results showed that: (1) under different atmosphere, the 2D-C/SiC composites subjected to thermal cycling behaved very differently and the most sensitive atmosphere was the wet oxygen; (2) external load could accelerate the degradation of the composites and changed the oxidation regimes of fibers; (3) the electrical resistance of the specimen could be detected on-line, stored in real time and analyzed reliably by the newly-developed SDIA; (4) 2D-C/SiC composites had an excellent thermal fatigue resistance in different environments.

Keywords: Carbon composites; Chemical vapor infiltration; Mechanical properties; Thermal expansion

1. Introduction

Carbon-fiber-reinforced SiC-matrix composites (C/SiC) fabricated by the chemical vapor infiltration process (CVI) have been proposed as advanced materials suitable for aerospace and gas turbine engine parts [1,2]. In particular, in recent years many efforts have been devoted to the high temperature applications of C/SiC composites. These composites show some attractive properties and advantages over traditional ceramics: higher tensile and flexural strength, enhanced fracture toughness and impact resistance, lower density and no cooling requirement. Especially, the mechanical properties of C/SiC composites can be retained at high temperatures and under severe service environments. However, the effects of thermal shock and thermal cycling on the composites have been anticipated to be an important factor and have resulted in degradation of performance in many instances. Consequently, thermal fatigue damage must be understood well before actual use.

Thermal shock resistance of monolithic materials has been extensively studied, and some theoretical analyses have been successfully applied to explain experimental observations [3–5]. Experimental thermal shock studies have been conducted on unidirectional, two-dimensional woven-fiber composites [6,7] and a newly-developed ascending thermal shock test equipment has also been designed to study thermal shock and thermal fatigue of ceramic materials [8]. However, a comprehensive understanding of the thermal fatigue behavior of 2D-carbon-fiber-reinforced ceramic composites (CFCCs) in different environments has not been obtained, despite

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recent advances in the architecture design and processing of these materials.

When evaluating CFCCs for potential use in cyclic temperature structural applications, the basic characterization of materials obtained from mechanical and environmental testing is very important in understanding the fundamental properties of the materials. In the present study, the behavior of a 2D-C/SiC composite (produced by CVI) subjected to thermal cycling in controlled environments was investigated completely and the results of thermal fatigue experiments in different environmental atmosphere are presented. The electrical resistance of 2D-C/SiC composite specimens in testing was acquired in real time by a newly-developed system of damage information acquisition (SDIA).

2. Experimental

2.1. Preparation of 2D-C/SiC composite

T-300™ carbon fiber from Toray (Japan) was employed. The fiber preform was prepared using a layered carbon-cloth braid method, and was supplied by the Nanjing Institute of Glass Fiber, People's Republic of China. The volume fraction of fibers was about 40%. Low pressure I–CVI was employed to deposit a pyrolytic carbon layer and the silicon carbide matrix. A thin pyrolytic carbon layer was deposited on the surface of the carbon fiber as the interfacial layer with C_3H_8 at 800 °C. Methyltrichlorosilane (MTS, CH_3 $SiCl_3$) was used for the deposition of the SiC matrix. MTS vapor was carried by bubbling hydrogen. Typical conditions for deposition were 1000 °C, a hydrogen: MTS ratio of $\alpha = 10$, and a pressure of 5 kPa. Argon was employed as the diluent gas to slow down the chemical reaction rate of deposition. Finally, the test specimens were machined from the fabricated composites and further coated with SiC (about 50 μm) by I–CVI under the same conditions. The dimension of the as-received C/SiC composite specimens is 3 mm × 3 mm × 185 mm. The virgin properties of the 2D-C/SiC composites are listed in Table 1.

2.2. Thermal shock tests

Thermal fatigue tests were conducted with a specific system including a high frequency induction heating furnace and a servo-hydraulic machine (Model INSTRON 8801 from INSTRON Ltd., England). The temperature was measured by an infrared pyrometer through a small window in the wall of the furnace and the wall was internally cut out to enable the circulating cold water to reach all over its surfaces. Thermal cycling was carried out between two selected temperatures by a programmable microprocessor and the period was 120 s (holding for 30 s at the lower temperature (less than 700 °C), heating to 1200 °C in 60 s and holding for 30 s, and then cooling back to the lower temperature immediately). The temperature difference ΔT was about 500 °C. Only the middle parts of specimens (about 40 mm long, 3 mm wide and 3 mm thick) were kept in the hot zone and were on-line quenched in environmental atmosphere, which included argon($\geqslant$99.99% and 1.01×10^5 Pa), dry oxygen (8000 Pa), water vapor (15 000 Pa, about 54 °C) and wet oxygen (coupled O_2 with H_2O pressures mentioned above). The flux of gases was accurately controlled by a mass flow controller (5850 i series of BROOKS, Japan) with a precision of 0.1 sccm. In testing, the loading mode was tension–tension fatigue (sine wave, fatigue stress level was about 50% of the as-received strength, frequency 1 Hz and stress ratio $R = 0.5$).

2.3. System of damage information acquisition for C/SiC composites

Changes in resistance can be caused by a wide range of mechanisms such as fiber breakage, delamination, mechanical strain and cyclic temperature. The system of damage information acquisition (SDIA) was designed to real time acquire the electrical resistance of the C/SiC composites subjected to thermal cycling in controlled environments by a sound card of personal computer. It comprised of the following major parts:

(1) Data Acquisition. As we know, the voltage signals changed with the electrical resistance of the vibrating carbon particles in microphones can be transformed into the digital signals by the sound card in the multimedia computer. Similarly, the voltage signals on both ends of the composite specimens in testing collected by the sound card reflect the electrical resistance of the specimens directly. Consequently, the data acquisition was developed

Table 1
Properties of the as-received 2D-C/SiC composites

Property	Density ($\times 10^3$ kg/m^3)	Modulus (GPa)	Strength (MPa)	Poisson's ratio	Porosity (%)	CTE* ($\times 10^{-6}$/°C)			
						600 °C	800 °C	1000 °C	1200 °C
Value	2.0	70	248	0.32	13	4.6	6.1	5.2	5.4

* CTE (the coefficient of thermal expansion) is measured by a dilatometer (Model DIL402C from NETSZCH, Germany).

according to the following steps: (a) Voltage signals at both ends of the composite specimens were detected and sent to a sound card by a sensor and STP with amplifier. (b) Data were acquired in real time, transformed into the digital signals, and then stored in the data acquisition engine of Matlab® [9] transitorily. (c) After the acquisition for 10 s, data were saved in the hard disc as the real-time resistance vs. time. The above steps (a)–(c) were repeated until specified durations of acquisition were completed or until the failure of the specimen, whichever was earlier. The specimen rate was set to 8 kHz and the interval of a frame was 10 ms. The calculation formula of the real-time resistance was given by,

$$R(\mathrm{i}) = 20C\sum_{n=1}^{N}\log_{10}u_{\mathrm{i}}^{2}(n) \tag{1}$$

$$R_{\mathrm{total}} = \int_{0}^{\infty} R(\mathrm{i})\cdot d\mathrm{i} \tag{2}$$

where, C is a transformation coefficient, R(i) is the real-time resistance of the frame i and its unit is dB, N the specimen number of the frame i, $u_{\mathrm{i}}(n)$ voltage of the specimen point n in the frame i, and R_{total} accumulated total resistance (an integral of R(i) vs. time di).

(2) Data Analyzer. This programme module could reload and analyze the recorded data by the functions in Matlab®, then drew diagrams in the windows or created reports directly.

2.4. Measurements and observations

The residual ultimate tensile strength of the specimen after thermal fatigues was measured on an INSTRON-8801 device at room temperature. The fracture surfaces were observed with a scanning electron microscope (SEM, HITACHI S-2700).

3. Results and discussion

3.1. Effect of thermal cycling in the different atmosphere on the mechanical properties

The sudden change in the surrounding temperature generates a temperature gradient, thereby, the ceramic body experiences thermal stress. The thermal stress created due to the temperature gradient is given by [10],

$$\sigma_{\mathrm{t}} = \frac{\alpha E \Delta T_{\mathrm{C}}}{1-\nu} \tag{3}$$

where σ_{t} is the thermal stress, α the coefficient of thermal expansion, E the Young's modulus, ΔT_{C} the critical temperature gradient and ν the Poisson's ratio.

In the present paper, according to the Table 1 the tensile strength of the as-received 2D-C/SiC was about 248 MPa, the value of υ approximated to 0.32, E was about 70 GPa and α was about 5.3×10^{-6}/°C on average. Therefore, from Eq. (3), we obtained: $\Delta T_{\mathrm{C}} \approx$ 455 °C, which was slightly lower than the temperature difference in testing ($\Delta T \approx 500$ °C). The thermal stress produced more and more cracks on the surfaces of the ceramic coating, and then propagated inwards when it exceeded the strength of the matrix material ($\Delta T > \Delta T_{\mathrm{C}}$).

The assessments of thermal fatigue damage in the different atmosphere were done by comparing the tensile stress vs. strain curves and the relative ultimate tensile strengths of 2D-C/SiC composites. The influence of thermal cycling (50 thermal cycles) on the mechanical properties in the different atmosphere is shown in Fig. 1. It is apparent from these curves in Fig. 1(a) that the elastic modulus and fracture strength were all lower for quenched composites than unquenched because of the lower slope of the linear portion of the curve and earlier deviation from linearity, respectively. The elastic modulus ascended continuously for original unquenched composites, however, for the quenched composites, it kept almost a steady value and the value was lower and lower with the following order of atmosphere:

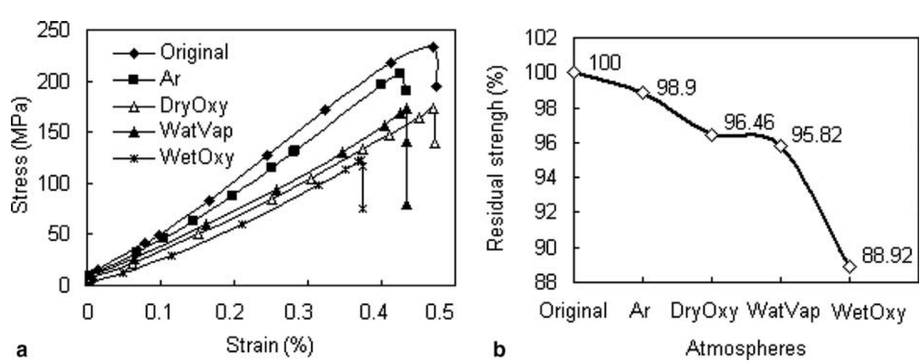

Fig. 1. (a) Typical stress vs. strain curves and (b) statistical results of residual strength in the argon, dry oxygen, water vapor and wet oxygen atmosphere (compared with original strength which was unquenched) after 50 thermal cycles.

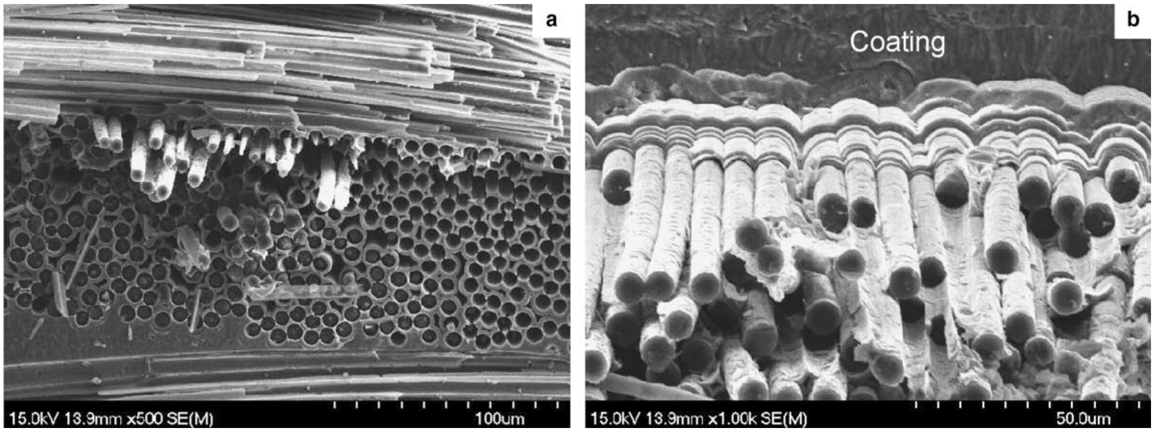

Fig. 2. Typical micrographs of the fracture sections of the 2D-C/SiC composites after 50 thermal cycles in wet oxygen atmosphere.

argon, dry oxygen, water vapor and wet oxygen. In addition, the residual strength (as shown in Fig. 1(b)), and work-of-fracture (area under the curve in Fig. 1(a)) also decreased according to the similar atmosphere order after 50 thermal cycles, compared with those of the original materials which were unquenched.

The mechanical properties of the composites were not only dependent on the environmental atmosphere, but also on the thermal cycling. The typical micrographs of the fracture sections of the 2D-C/SiC composites after 50 thermal cycles in the wet oxygen atmosphere are presented in Fig. 2. It can be seen that the fibers were oxidized and/or pulled out and that the opening transverse cracks of SiC matrix and matrix delaminations were very regular (almost equal intervals) due to thermal mismatch from exterior to interior and cyclic thermal stress. High strength fibers hindered the propagation of matrix cracks, and the mechanical degradation was reduced by crack deflection and crack bridging. Therefore, the bonding between fibers and matrix should be weak enough in order to make the fibers and matrix debond and the cracks deflect around fibers. Otherwise, the matrix cracks would propagate into the fibers. The effects of thermal cycling in the oxidizing atmosphere on the maximum tensile strength could result from the following two causes: (i) the oxidation of the fibers in the corrosive gas and (ii) the fracture of fibers under the thermal stress and/or the interface damage due to thermal mismatch. The decrease of modulus of composites could be ascribed to matrix cracking, fiber fracture and interface debonding.

3.2. The most sensitive atmosphere to 2D-C/SiC composites under the thermal fatigue was wet oxygen

When C/SiC composites were cooled from high to low temperature, tensile stress on the surface and compressive stress in the core were created by the thermal gradient, which caused a large quantity of matrix cracks on the surface of composites. These micro-crack fissures provide paths through which oxygen could migrate toward the carbon reinforcement and reacted with it. In the present study, however, carbon fibers were hardly oxidized by dry oxygen after 50 thermal cycles (as shown in Fig. 3(a)). A dense protective oxidation film (silica), too short period of exposure and self-healing microcracks of the coating in thermal fatigue testing were responsible for this result. The behaviour of C/SiC in

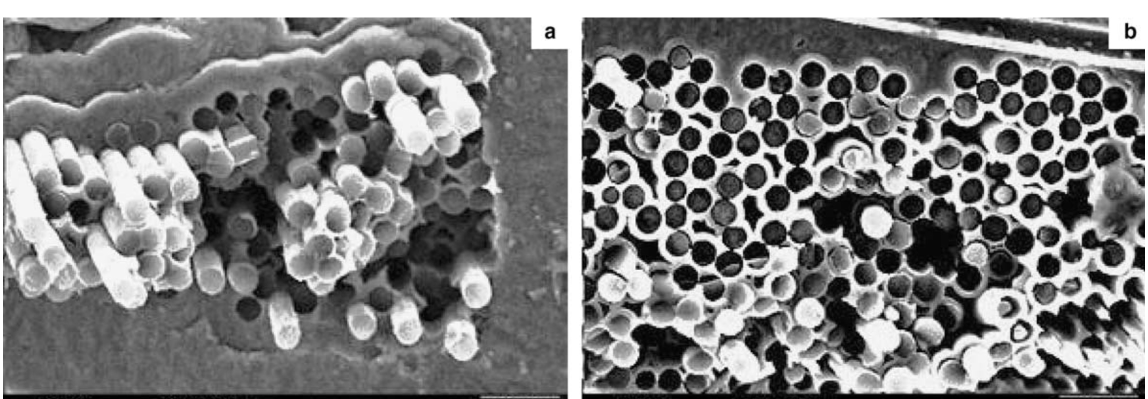

Fig. 3. SEM micrograph of fracture section of 2D-C/SiC in (a) dry oxygen and (b) wet oxygen atmosphere after 50 thermal cycles.

dry oxygen has been well studied. The SiC matrix acts as an oxygen barrier for the carbon fiber due to the formation of SiO_2 during its oxidation. In testing, the temperature difference ($\Delta T \approx 500$ °C) was slightly higher than the critical value ($\Delta T_C \approx 455$ °C), thermal stress was not enough for crack opening completely. On the one hand, long oxygen diffusion paths (as shown in Fig. 2(b)), the matrix delaminations and crack deflection and bridging provided a longer gas diffusion path.), incomplete open cracks and dense silica layers made oxidation of the fibers difficult in a short time of exposure once cooling. On the other hand, when the temperature was above 900 °C, self-healing of cracks due to thermal expansion and sealing of crack due to amorphous silica made oxidation of the internal fibers more difficult.

However, damage and degradation of the fibers was very severe in the oxygen containing water vapor (wet oxygen) after the same 50 thermal cycles. The typical morphology of the fracture section of 2D-C/SiC in the dry oxygen and wet oxygen after 50 thermal cycles is presented in Fig. 3. It appears that the wet oxygen atmosphere is more corrosive than a dry one to CVD–SiC matrix covering the carbon fibers and the fibers are subjected to a superficial oxidation. There is a general agreement within the literature that oxygen and water vapor enhances the oxidation rate of SiC in the passive regime according to the following reactions:

$$SiC_{(s)} + 3/2O_{2(g)} \rightarrow SiO_{2(s)} + CO_{(g)} \tag{4}$$

$$SiC_{(s)} + 3H_2O_{(g)} \rightarrow SiO_{2(s)} + 3H_{2(g)} + CO_{(g)} \tag{5}$$

Water vapor plays a strong influence on the crystallization of amorphous silica which could accelerate SiC oxidation by promoting devitrification. Crystallized silica could not act as an oxygen barrier by sealing the cracks under cyclic temperature. In addition, the dense silica layer was damaged by water. Water is dissolved as molecules in the amorphous silica layer and then reacts with the silicon–oxygen lattice to form Si–OH bonds [11]. The formation of hydroxylic groups into the silica network due to the presence of water vapor in an oxygen stream produces a less dense silica film which allows for faster diffusion of oxidizing species and/or reaction products [12]. Consequently, the role playing by SiC as an oxidation barrier seems to be limited although its oxidation is noticeable in H_2O. The most sensitive atmosphere to 2D-C/SiC composites under the thermal cycling was wet oxygen.

3.3. Fatigue load could force cracks of matrix to open up and close up periodically, and caused the uniform oxidation of 2D-C/SiC composites

As mentioned above, fibers were hardly oxidized under the thermal fatigue in dry oxygen atmosphere without external load. However, in coupled environments of dry oxygen and fatigue load, fatigue stress could force cracks of matrix to open up and close up along the axial direction of fiber bundles (0–90° woven C/SiC) periodically. Therefore, surface cracks propagated inwards and crossed each other easily because only halves of the fibers in composites (longitudinal fibers) can hinder the propagation of cracks. As we know, cracks perpendicular to the fibers are more difficult to penetrate through the fiber bundles and propagate into the interiors of the specimens than cracks parallel to the fibers. Under cyclic loading, transverse coating cracks in 2D specimens easily penetrate through the transverse (90°) fiber bundles and propagate inwards. Oxygen channels were formed around the longitudinal fibers and caused a uniform oxidation. As shown in Fig. 4, in the interior of the 2D-C/SiC specimen, fibers became thinner and thinner after oxidation around them uniformly. External load accelerated the degradation of composites and changed the oxidation regimes of fibers.

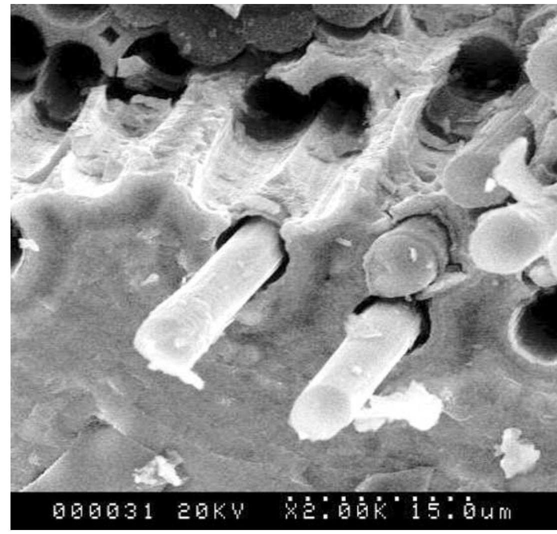

Fig. 4. SEM micrograph of fracture section of 2D-C/SiC under coupled environments of dry oxygen and fatigue load.

3.4. The thermal fatigue damage to 2D-C/SiC composites was limited and there existed a critical thermal shock number

Degradation of mechanical properties of C/SiC under the thermal shock was due to a physical damage, not a chemical corrosion (e.g. oxidation). When the quenching number was larger than a critical value, the crack density was saturated, and the strength of the composites did not decrease any further [13] (for 3D C/SiC composites). Therefore, damage of thermal cycling, actually, could be understood as a destructive energy to composites. The energy absorbed into materials caused physical damage of materials (such as matrix cracking, fiber fracture and interface debonding) in order to release itself

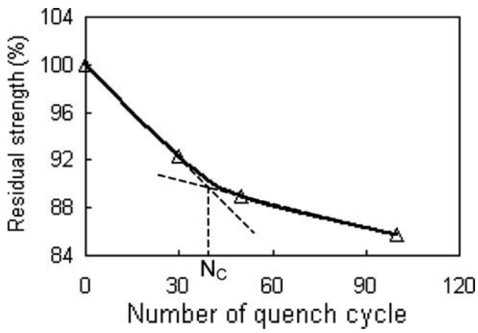

Fig. 5. Effect of thermal shock number on the residual strength of 2D-C/SiC in wet oxygen atmosphere.

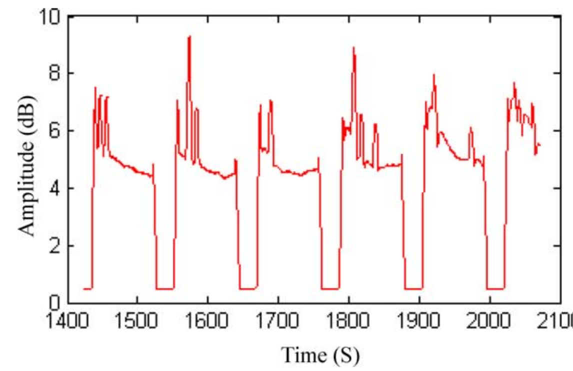

Fig. 6. Time-domain analysis of typical signals acquired by SDIA under the thermal cycling in wet oxygen atmosphere.

for thermal balance. When damage to materials was saturated, materials had not any response to the destructive energy of thermal cycling because macro-cracks of matrix had enough space to tolerate micro-thermal-expansion. Accordingly, the damage-saturated C/SiC composite above the critical thermal cycles could not absorb any more destructive energy. In other words, the thermal fatigue damage to 2D-C/SiC composites was also limited.

As shown in Fig. 5, the statistical result of the thermal fatigue tests in the wet oxygen atmosphere showed that the residual strength of 2D-C/SiC composites decreased gradually with increasing cycle number and the decrement was less and less. An inflexion occurred on the curve when the critical number N_C was about 40. After this point, the continuous and slow decrease of the mechanical properties should be ascribed to oxidation of fibers in wet oxygen with increasing quenching time.

3.5. *The critical number of thermal cycles of 2D-C/SiC composites could be reliably forecasted by the SDIA*

The damage revealed by electrical resistance measurement was subtle, in contrast to damage in the form of well-defined delamination cracks, fiber breakage or debonded regions in composites. Therefore, correlation of the resistance change with microscopic observation of damage was impossible. The law of damage, however, could be obtained by real time monitoring of resistance using the newly-developed system of damage information acquisition.

The amplitude of the voltage signal resulting from vibrations of the specimen resistance under thermal cycling was displayed in real time on the screen (Fig. 6). Voltage signals changed periodically with thermal cycles and their periods were equal approximatively (about 120 s). The vibrations of voltage signals reflected changes in resistance of the composite specimen directly. It was more interesting that resistance decreased upon heating and increased with cooling in each cycle. As expected, the greater the damage in cooling, the more severe are the spikes or resistance increase. It is not surprising that the resistance of the composites is rather larger in cooling than upon heating because fibers are broken and parted from each other once cooling. In addition, the damage was most significant in the early cycles and incremental damage diminished upon thermal cycling. A typical relative resistance change ($\Delta R/R_0$) vs. time curve of the specimen in the wet oxygen atmosphere is presented in Fig. 7 (left). During the initial stage of thermal cycling, with increasing quench number, surface cracks propagated inwards, which led to the increase in crack density of the composites. The increase in matrix crack density resulted in the rapid decrease of the mechanical properties of the composites. According to Fig. 7 (left), the energy absorbed into the specimen was very high and rapidly dissipated to destroy the materials at the first stage. Subsequently, minor damage occurred gradually as cycling progressed, leading to the gradual and irreversible decrease of the baseline resistance. Finally, it also remained a lower steady level at about 4500 s, which was consistent with the above experimental statistical results (about 40 cy-

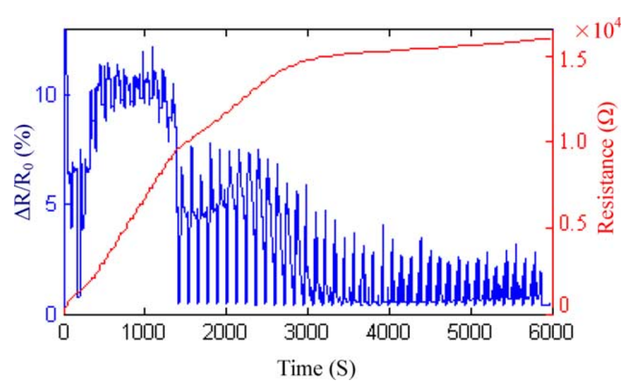

Fig. 7. Typical curve of $\Delta R/R_0$ vs. time recorded by SDIA (left) and accumulated resistance of specimen (right) under thermal cycling in the wet oxygen atmosphere.

cles according to Fig. 5). The accumulated resistance of the specimen (R_{total}) was a mathematic integral of the left curve in Fig. 7 according to formula (2) (the area under the left curve vs. time). The right curve in Fig. 7 shows that the integral curve approximated to an increasing parabola (power < 1) and finally became a low slope beeline. At this time, matrix cracks were saturated and thermal damage to the composites reached a maximum.

3.6. 2D-C/SiC composites had an excellent thermal fatigue resistance in different environments

As shown in Fig. 1(b) and Fig. 5, after the 2D-C/SiC composite was quenched 50 times in the wet oxygen atmosphere between 700 °C and 1200 °C, the residual strength of the composite retained 88.92% of the virgin strength, and retained 98.90%, 96.46%, 95.82% in the argon, dry oxygen and water vapor atmosphere, respectively. Furthermore, after the quenching number was more than 100 times in the wet oxygen, the residual tensile strength was also 86.69% of the virgin strength. Even under coupled environments of wet oxygen with fatigue load and after 100 thermal cycles, strength loss was still less. This should be partly ascribed to limited and short testing time (about 200 minutes for 100 thermal cycles). Consequently, 2D-C/SiC composites had an excellent thermal fatigue resistance in different environments.

4. Conclusions

1. Different atmosphere under thermal fatigues affected the mechanical properties of 2D-C/SiC composites differently. The elastic modulus, residual strength and work-of-fracture became lower and lower with the following order of atmosphere: argon, dry oxygen, water vapor and wet oxygen. The composites were the most sensitive to the wet oxygen.
2. External load could accelerate the degradation of composites and changed the oxidation regimes of fibers. Cyclic fatigue load could force cracks of matrix to open up periodically, as a result surface cracks propagated inwards and crossed each other easily, uniform oxidation occurred in 2D-C/SiC composites.
3. Thermal fatigue damage to 2D-C/SiC composites was limited and there existed a critical thermal cycling number. The critical number of 2D-C/SiC composites in the wet oxygen was about 40 times according to the experimental statistical results, which was consistent with that forecasted by the SDIA method (electrical resistance characterization). Thermal fatigue damage was saturated when the total resistance of composites reached maximum.
4. 2D-C/SiC composites had a good thermal fatigue resistance in different environments. After the composite was quenched more than 100 times in the severest wet oxygen atmosphere, the residual tensile strength was still 86.69% of the virgin strength.

Acknowledgements

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